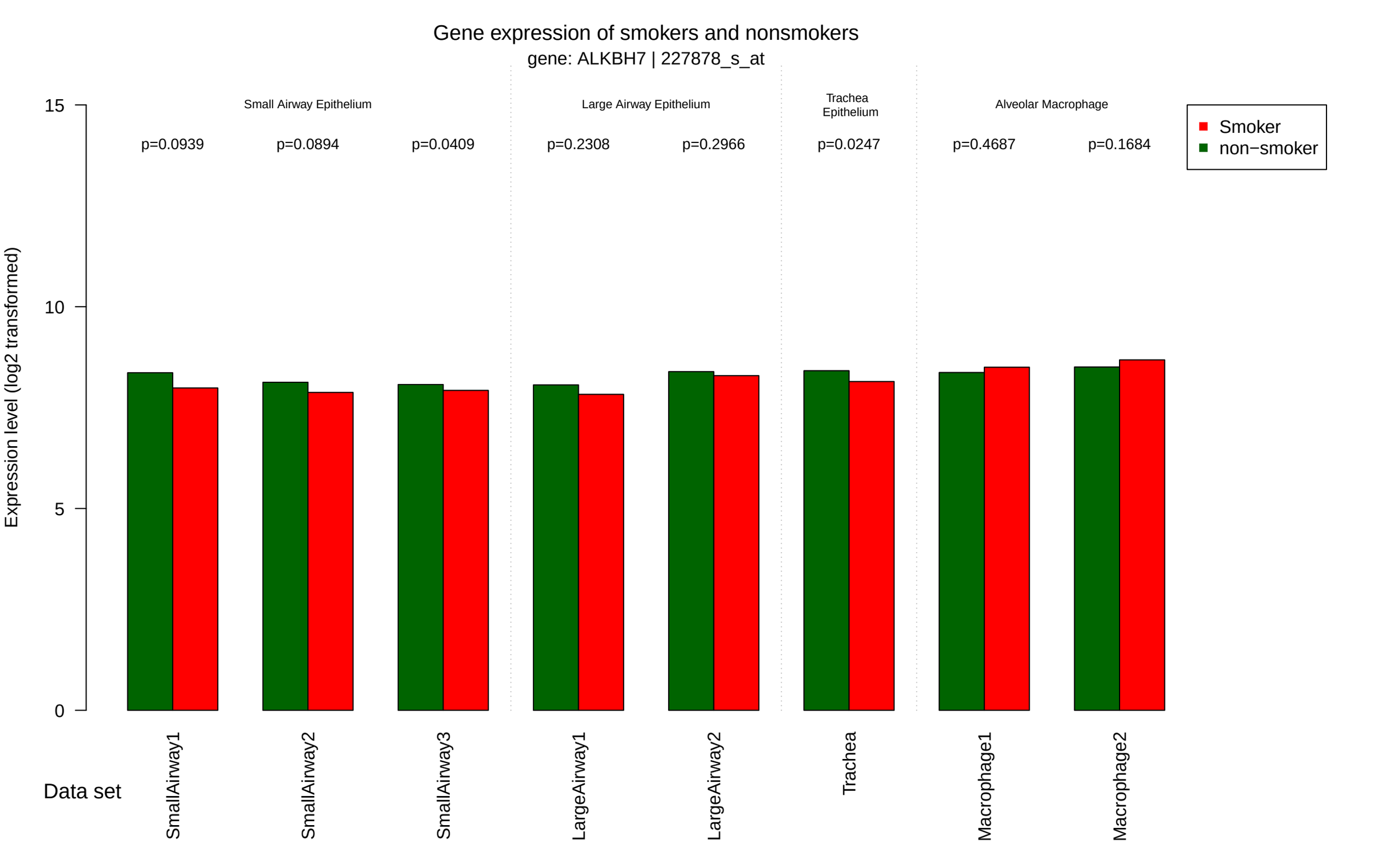
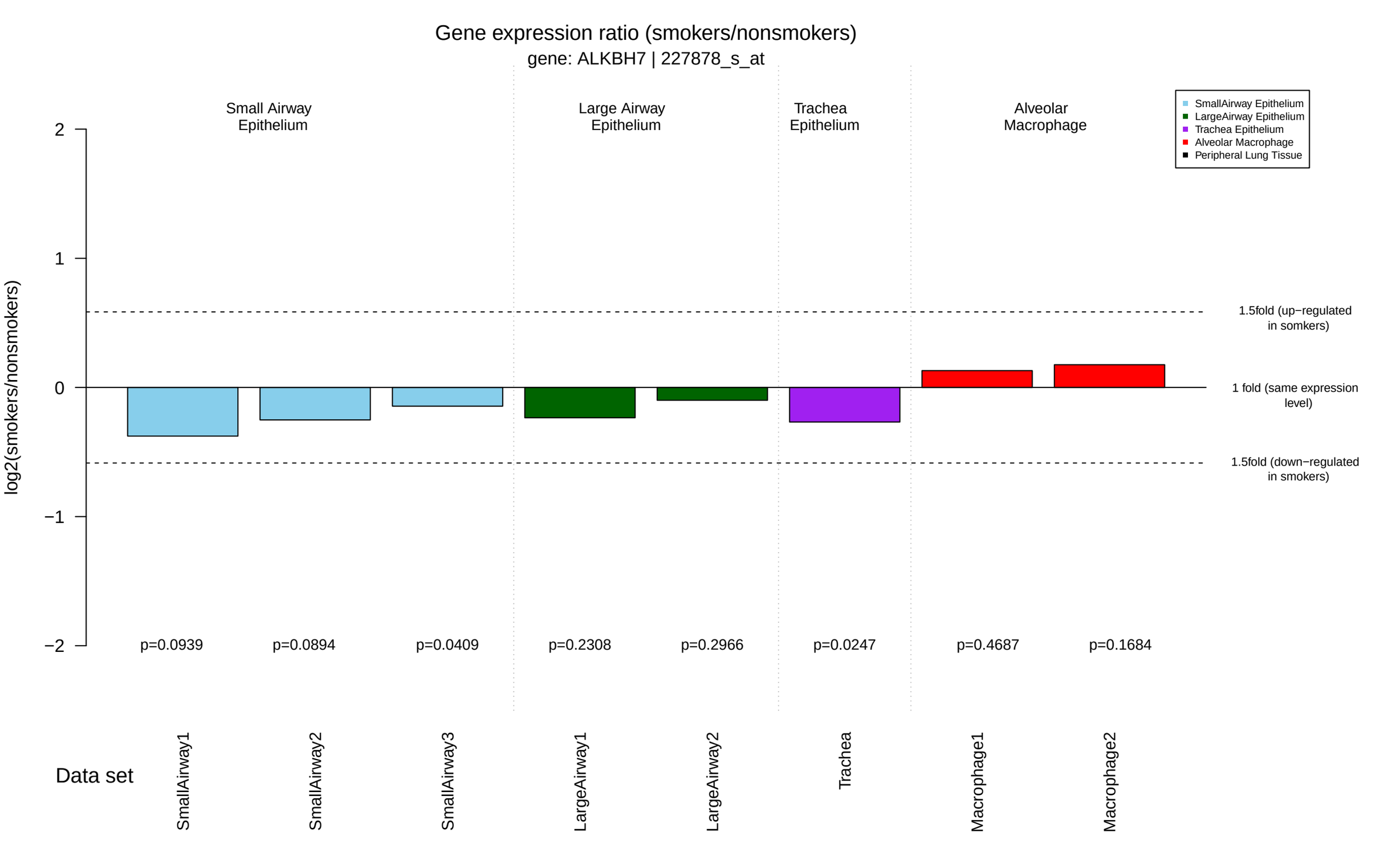
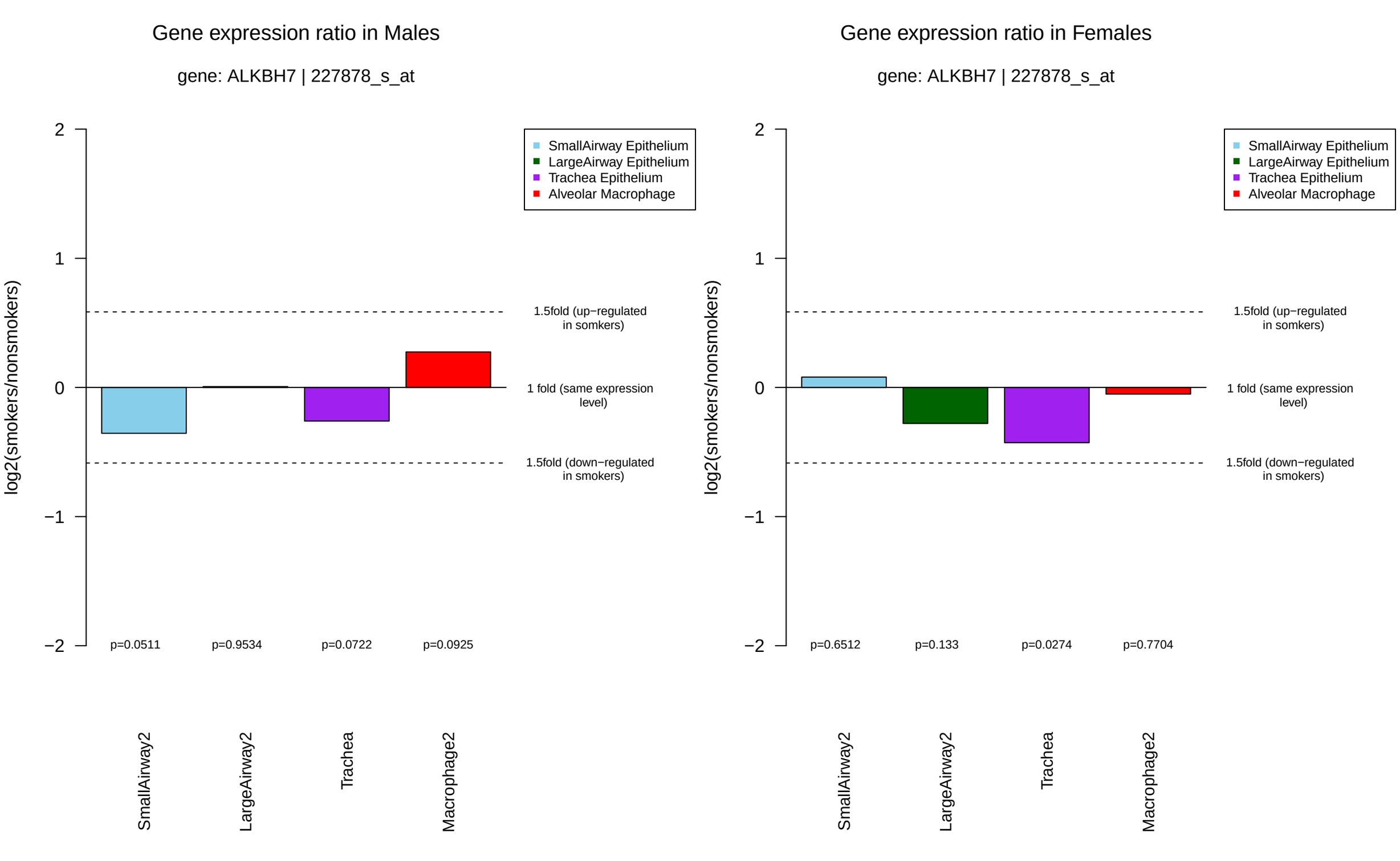
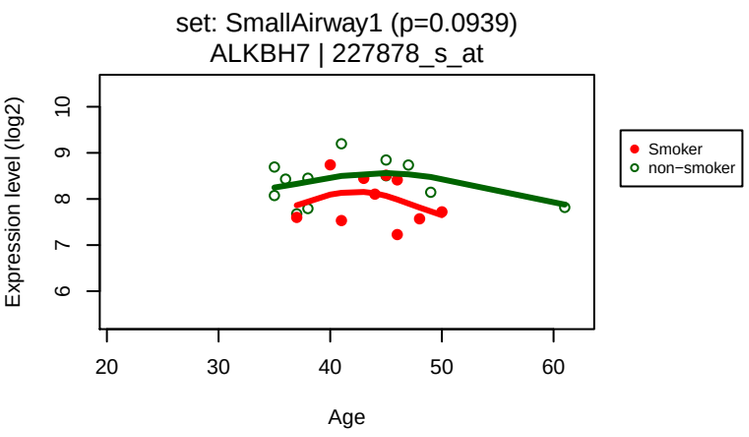
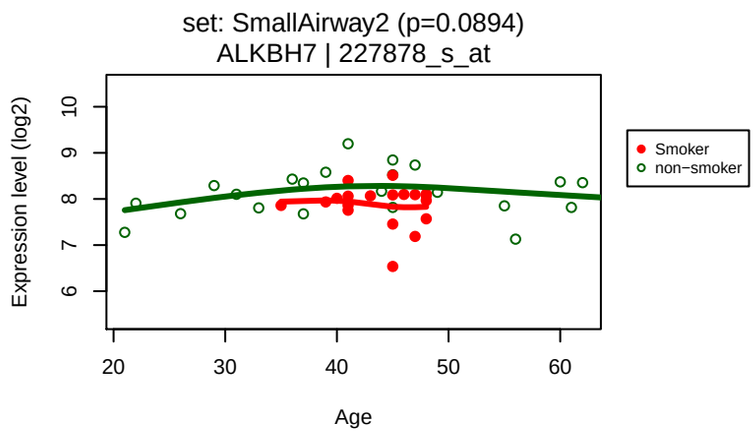
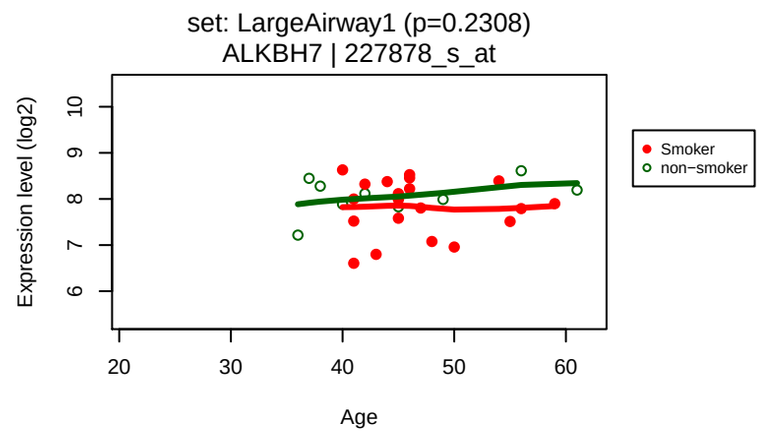

Gene expression of smokers and nonsmokers
gene: ALKBH7 | 227878_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0939
p=0.0894
p=0.0409
p=0.2308
p=0.2966
p=0.0247
p=0.4687
p=0.1684
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

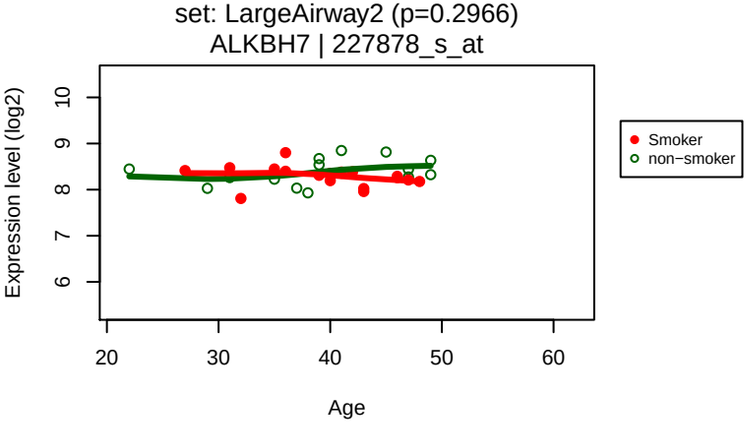

Gene expression ratio (smokers/nonsmokers)
gene: ALKBH7 | 227878_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0939
p=0.0894
p=0.0409
p=0.2308
p=0.2966
p=0.0247
p=0.4687
p=0.1684
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

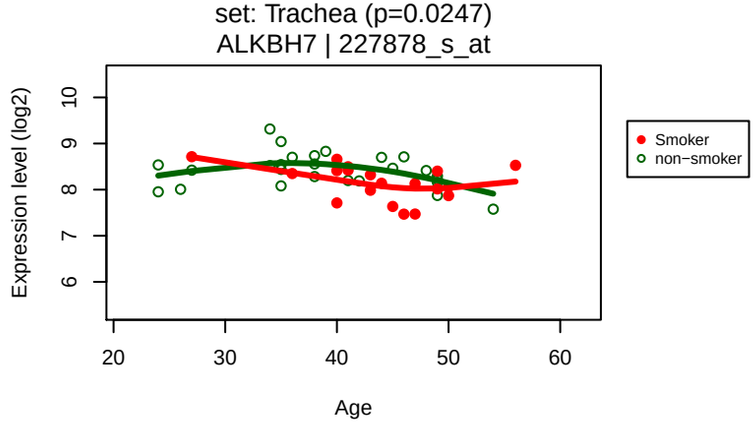

Gene expression ratio in Males
gene: ALKBH7 | 227878_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0511
p=0.9534
p=0.0722
p=0.0925
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALKBH7 | 227878_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6512
p=0.133
p=0.0274
p=0.7704
SmallAirway2
LargeAirway2
Trachea
Macrophage2

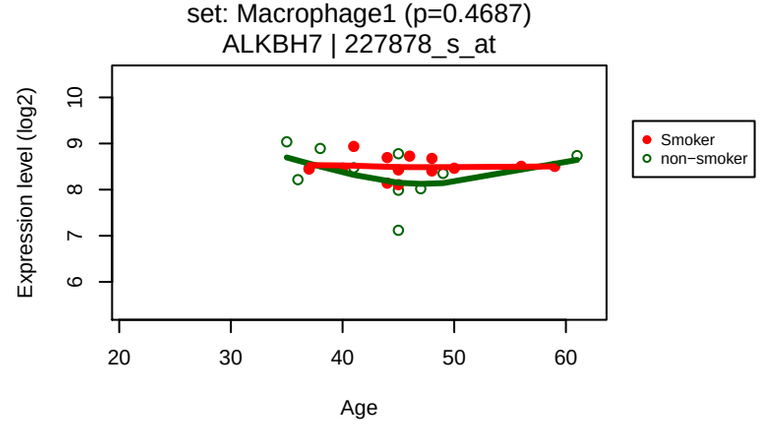

set: SmallAirway1 (p=0.0939)
ALKBH7 | 227878_s_at
Expression level (log2)
Age
Smoker
non−smoker

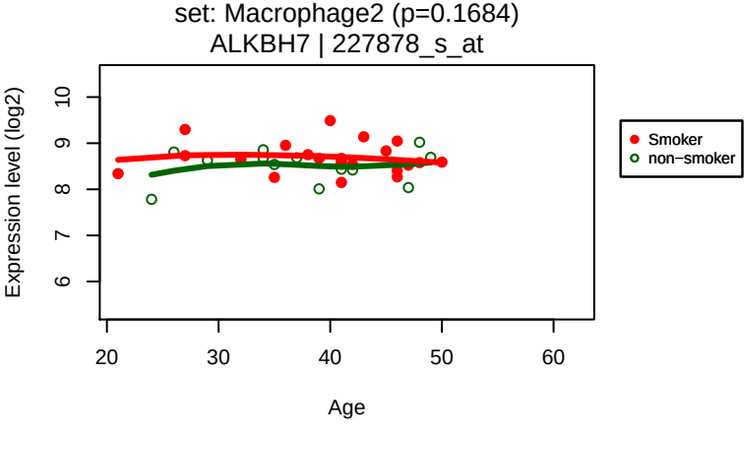

set: SmallAirway2 (p=0.0894)
ALKBH7 | 227878_s_at
Expression level (log2)
Age
Smoker
non−smoker

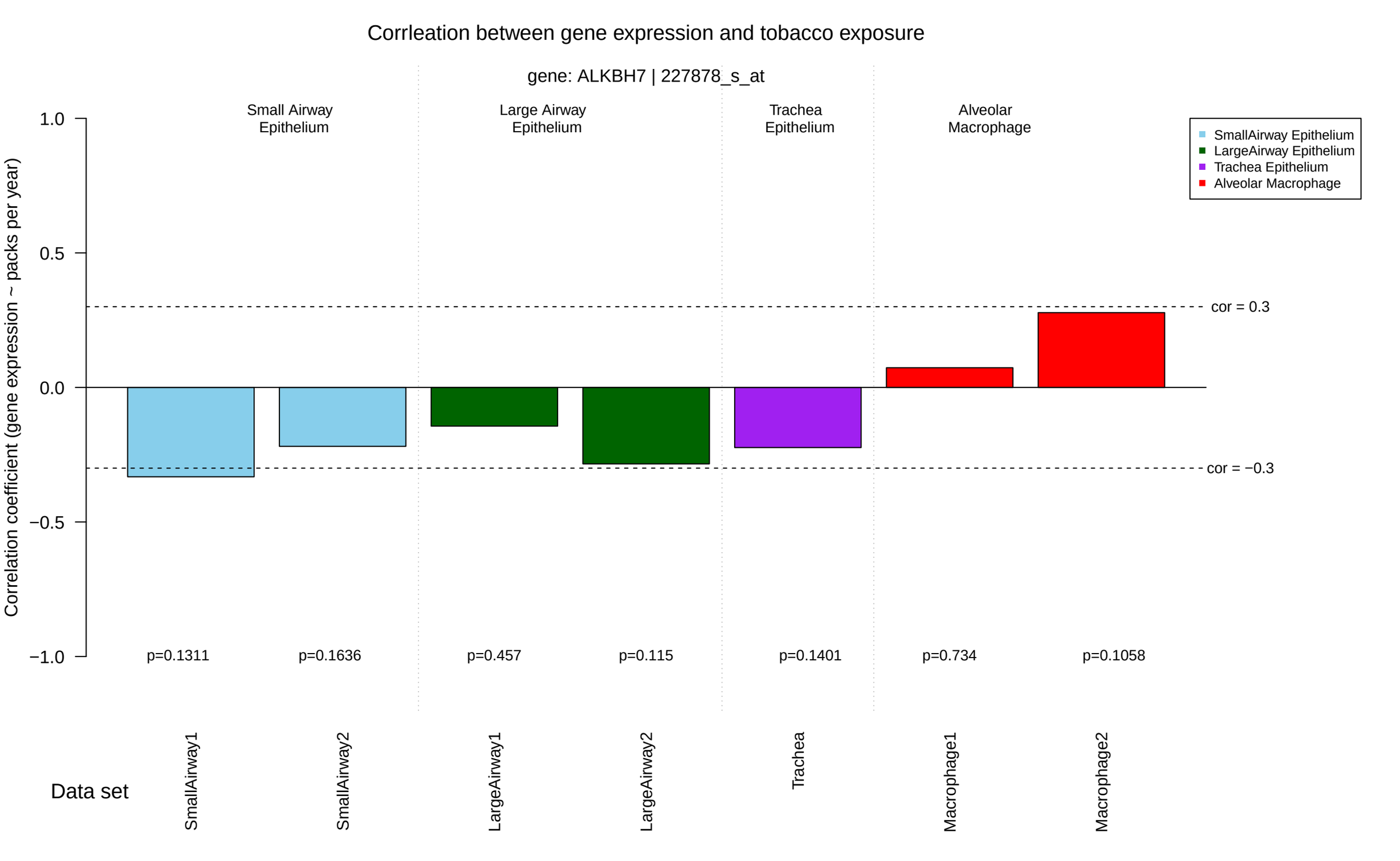

set: LargeAirway1 (p=0.2308)
ALKBH7 | 227878_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2966)
ALKBH7 | 227878_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0247)
ALKBH7 | 227878_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4687)
ALKBH7 | 227878_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1684)
ALKBH7 | 227878_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALKBH7 | 227878_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1311
p=0.1636
p=0.457
p=0.115
p=0.1401
p=0.734
p=0.1058
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2